

Prerequisites (Needed to Successfully Run Project)

Requirement	Use in this project
Java + Maven	Compiles and runs the SparkJava backend service
MySQL	Stores the data layer for ingredients, recipes, and meal prep records
Python simple server	Hosts the front-end files locally from the web-client folder
Browser	Opens the client and tests the running application
IDE (optional)	Can be IntelliJ, VS Code, or another Java IDE for easier editing

Download and Unpack the Project

1. Open the GitHub repository for the project.
2. Click the **Code** button and choose Download ZIP.
3. Unzip the project to a folder on your computer.
4. Open the project root in a terminal or IDE.

Set Up the Database

5. Start your local MySQL server.
6. In the project folder, open the sql directory.
7. Run the SQL files in this order: create_schema.sql, then fill_tables.sql.
8. Open the backend database utility file and confirm the MySQL username, password, and connection URL match your local machine.
9. If the database is not running, or the credentials are wrong, the frontend may still load, but CRUD requests will fail.

Build and Run the Back End

Open first terminal ****cd into project root****, run:

```
mvn clean compile
```

```
mvn -q exec:java "-Dexec.mainClass=MptApiServer"
```

How to verify success: Open <http://localhost:8080/health> in a browser. A successful launch should return OK.

Run the Front End

Open a second terminal ****cd into project root****, then run:

```
cd web-client
```

```
python -m http.server 5500
```

If python does not work, use `python3 -m http.server 5500`

Open the client at: <http://localhost:5500/>

How the Layers Are Hosted

- Client layer: hosted locally by the Python simple server from the web-client folder.
- Service layer: hosted locally by the Java Spark server.
- Business layer: runs inside the Java backend.
- Data layer: stored in MySQL and accessed only through the service layer.

How to Identify a Successful Setup

- The backend starts without database connection errors.
- The <http://localhost:8080/health> endpoint returns OK.
- The browser opens the front end successfully.
- Creating, getting, and updating data work from the UI.
- The database tables reflect the same changes after requests are made.

Common Problems

Problem	What to check
Backend fails to start	Check that MySQL is running and the DB credentials are correct.
Health page does not return OK	Make sure the Java server is still running in the terminal.
Frontend opens but data actions fail	This usually means the backend is not running or cannot connect to MySQL.
No page at localhost:5500	Restart the Python simple server from the web-client folder.

Note: The SQL files used in this project are named `create_schema.sql` and `fill_tables.sql`.